

```
create database shubham;

use shubham;

CREATE TABLE Employees (
EmpID INT PRIMARY KEY,
Name VARCHAR(50) NOT NULL,
Dept VARCHAR(20) NOT NULL,
Role VARCHAR(20) NOT NULL,
Salary INT NOT NULL,
Gender CHAR(1) NOT NULL,
JoinDate DATE NOT NULL,
Address VARCHAR(100) NOT NULL,
City VARCHAR(50) NOT NULL,
State CHAR(2) NOT NULL,
Region VARCHAR(10) NOT NULL
);

INSERT INTO Employees (EmpID, Name, Dept, Role, Salary, Gender, JoinDate,
Address, City, State,
Region) VALUES
(1, 'Alice', 'HR', 'Manager', 55000, 'F', '2019-03-15', '101 Maple St', 'New York', 'NY',
'East'),
(2, 'Bob', 'IT', 'Developer', 72000, 'M', '2020-07-22', '22 Oak Ave', 'Chicago', 'IL',
'Midwest'),
(3, 'Carol', 'Finance', 'Analyst', 68000, 'F', '2018-11-01', '303 Pine Rd', 'New York',
'NY',
'East'),
(4, 'David', 'HR', 'Executive', 52000, 'M', '2021-01-10', '44 Birch Ln', 'Chicago', 'IL',
'Midwest'),
(5, 'Eve', 'IT', 'Senior Dev', 85000, 'F', '2017-06-30', '55 Cedar Blvd', 'Houston', 'TX',
'South'),
```

(6, 'Frank', 'Finance', 'Manager', 74000, 'M', '2019-09-14', '606 Elm St', 'New York', 'NY', 'East'),

(7, 'Grace', 'Marketing', 'Executive', 60000, 'F', '2022-02-28', '77 Spruce Dr', 'Chicago', 'IL', 'Midwest'),

(8, 'Hank', 'IT', 'Lead', 90000, 'M', '2016-04-05', '88 Walnut Way', 'Houston', 'TX', 'South'),

(9, 'Ivy', 'HR', 'Executive', 58000, 'F', '2020-12-19', '99 Ash Ct', 'New York', 'NY', 'East'),

(10, 'Jack', 'Finance', 'Analyst', 65000, 'M', '2021-03-07', '10 Willow Pl', 'Chicago', 'IL', 'Midwest'),

(11, 'Karen', 'Marketing', 'Manager', 62000, 'F', '2018-08-23', '11 Poplar St', 'Houston', 'TX', 'South'),

(12, 'Leo', 'IT', 'Developer', 78000, 'M', '2019-11-11', '12 Sycamore Ave', 'New York', 'NY', 'East'),

(13, 'Mia', 'HR', 'Executive', 53000, 'F', '2022-05-16', '13 Hickory Rd', 'Chicago', 'IL', 'Midwest'),

(14, 'Nate', 'Finance', 'Manager', 71000, 'M', '2017-12-02', '14 Chestnut Ln', 'Houston', 'TX', 'South'),

(15, 'Olivia', 'Marketing', 'Analyst', 66000, 'F', '2020-06-18', '15 Magnolia Dr', 'New York', 'NY', 'East'),

(16, 'Paul', 'IT', 'Senior Dev', 82000, 'M', '2018-03-27', '16 Dogwood Blvd', 'Chicago', 'IL', 'Midwest'),

(17, 'Quinn', 'HR', 'Executive', 57000, 'F', '2021-09-09', '17 Hawthorn St', 'Houston', 'TX',

'South'),
(18, 'Rita', 'Finance', 'Lead', 76000, 'F', '2019-01-25', '18 Redwood Ave', 'New York',
'NY',
'East'),
(19, 'Sam', 'Marketing', 'Executive', 63000, 'M', '2022-07-14', '19 Cypress Ct',
'Chicago', 'IL',
'Midwest'),
(20, 'Tina', 'IT', 'Lead', 88000, 'F', '2016-10-31', '20 Juniper Way', 'Houston', 'TX',
'South'),
(21, 'Uma', 'HR', 'Manager', 60000, 'F', '2020-04-03', '21 Aspen Pl', 'Dallas', 'TX',
'South'),
(22, 'Victor', 'Finance', 'Analyst', 69000, 'M', '2018-07-19', '22 Cottonwood St', 'Dallas',
'TX',
'South'),
(23, 'Wendy', 'Marketing', 'Manager', 64000, 'F', '2021-11-27', '23 Palmetto Ave',
'Phoenix',
'AZ', 'West'),
(24, 'Xander', 'IT', 'Architect', 95000, 'M', '2015-02-14', '24 Mesquite Rd', 'Phoenix',
'AZ',
'West'),
(25, 'Yara', 'HR', 'Executive', 54000, 'F', '2022-09-08', '25 Saguaro Ln', 'Phoenix', 'AZ',
'West'),
(26, 'Zoe', 'Finance', 'Manager', 73000, 'F', '2019-05-21', '26 Palo Verde Dr', 'Dallas',
'TX',
'South'),
(27, 'Aaron', 'Marketing', 'Analyst', 67000, 'M', '2020-10-15', '27 Ocotillo Blvd',
'Phoenix',
'AZ', 'West'),
(28, 'Bella', 'IT', 'Senior Dev', 80000, 'F', '2017-08-06', '28 Ironwood St', 'Chicago', 'IL',
'Midwest'),

(29, 'Carlos', 'HR', 'Executive', 56000, 'M', '2021-06-24', '29 Pecan Ave', 'Houston', 'TX', 'South'),

(30, 'Diana', 'Finance', 'Lead', 77000, 'F', '2018-02-09', '30 Peachtree Ct', 'New York', 'NY', 'East'),

(31, 'Ethan', 'IT', 'Developer', 74000, 'M', '2019-04-18', '31 Magnolia St', 'Dallas', 'TX', 'South'),

(32, 'Fiona', 'Marketing', 'Executive', 61000, 'F', '2022-01-05', '32 Bluebonnet Ave', 'Dallas', 'TX', 'South'),

(33, 'George', 'Finance', 'Analyst', 66000, 'M', '2020-08-11', '33 Sunflower Rd', 'Phoenix', 'AZ', 'West'),

(34, 'Helen', 'HR', 'Manager', 59000, 'F', '2017-03-22', '34 Cactus Ln', 'Phoenix', 'AZ', 'West'),

(35, 'Ivan', 'IT', 'Lead', 92000, 'M', '2015-09-30', '35 Desert Dr', 'Phoenix', 'AZ', 'West'),

(36, 'Julia', 'Marketing', 'Manager', 65000, 'F', '2019-12-03', '36 Mesa Blvd', 'Dallas', 'TX', 'South'),

(37, 'Kevin', 'Finance', 'Manager', 70000, 'M', '2021-07-20', '37 Prairie St', 'Chicago', 'IL', 'Midwest'),

(38, 'Laura', 'HR', 'Executive', 55000, 'F', '2022-10-14', '38 Lakeview Ave', 'Chicago', 'IL', 'Midwest'),

(39, 'Mark', 'IT', 'Developer', 76000, 'M', '2018-05-07', '39 Riverview Rd', 'New York', 'NY', 'East'),

(40, 'Nina', 'Marketing', 'Analyst', 63000, 'F', '2020-03-25', '40 Bayview Ln', 'New York', 'NY', 'East'),

(41, 'Oscar', 'Finance', 'Analyst', 67000, 'M', '2019-08-19', '41 Hillside Dr', 'Dallas', 'TX', 'South'),

(42, 'Priya', 'HR', 'Senior Dev', 79000, 'F', '2016-11-28', '42 Creekside Blvd', 'Houston', 'TX', 'South'),

(43, 'Raj', 'IT', 'Manager', 85000, 'M', '2018-06-14', '43 Brookside St', 'Phoenix', 'AZ', 'West'),

(44, 'Sarah', 'Marketing', 'Lead', 71000, 'F', '2017-02-01', '44 Springdale Ave', 'New York', 'NY', 'East'),

(45, 'Tom', 'Finance', 'Senior Dev', 81000, 'M', '2020-09-09', '45 Fairview Rd', 'Chicago', 'IL', 'Midwest'),

(46, 'Uma', 'IT', 'Architect', 97000, 'F', '2014-12-22', '46 Northgate Ln', 'New York', 'NY', 'East'),

(47, 'Vera', 'Marketing', 'Manager', 68000, 'F', '2021-04-30', '47 Eastwood Dr', 'Dallas', 'TX', 'South'),

(48, 'Will', 'HR', 'Analyst', 51000, 'M', '2022-11-15', '48 Westfield Blvd', 'Phoenix', 'AZ', 'West'),

(49, 'Xena', 'Finance', 'Lead', 78000, 'F', '2019-10-06', '49 Southpark St', 'Chicago', 'IL', 'Midwest'),

(50, 'Yusuf', 'IT', 'Senior Dev', 87000, 'M', '2016-07-17', '50 Central Ave', 'Houston', 'TX', 'South');

-- Q1=

select dept, count(*) as employees from employees group by dept;

-- Q2=

select dept, sum(salary) as totalsalary from employees group by dept;

-- Q3=

select gender, avg(salary) as avgragesalary from employees group by gender;

-- Q4=

select role, min(salary) as minimumsalary from employees group by role;

-- Q5=

select dept, max(salary) as maximumsalary from employees group by dept;

-- Q6=

select city, count(*) as countemployees from employees group by city;

-- Q7=

select region, sum(salary) as totalsalary from employees group by region;

-- Q8=

select dept, avg(salary) as averagesalary from employees group by dept;

-- Q9=

select city, sum(salary) as totalsalary from employees group by city;

-- Q10=

```
select role, count(name) as rolenamen from employees group by role;
```

```
-- Q11=
```

```
select State, count(*) as countemployees from employees group by State ORDER  
BY countemployees DESC, State ASC;
```

```
-- Q12=
```

```
select region, sum(salary) as totalsalary from employees group by region order by  
totalsalary DESC, region ASC;
```

```
-- Q13=
```

```
select dept, min(salary) as minimumsalary from employees group by dept order by  
minimumsalary desc, dept asc;
```

```
-- Q14=
```

```
select dept, count(*) as employees from employees group by dept ;
```

```
-- Q15
```

```
select dept, avg(salary) as employeeessalary from employees group by dept having  
avg(salary)> 65000;
```

```
-- Q16=
```

```
select city, sum(salary) as totalsalary from employees group by city having  
sum(salary)> 600000;
```

```
-- Q17=
```

```
select role, count(name) as employees from employees group by role having  
count(name)>7;
```

```
-- Q18=
```

```
select region, avg(salary) as averagesalary from employees group by region having  
avg(salary)<70000;
```

-- Q19=

```
select dept, max(salary) as maxmumsalary from employees group by dept having  
max(salary)>90000;
```

-- Q20=

```
select state, count(name) as headcount from employees group by state having  
count(name)>12;
```

-- Q21=

```
select city, min(salary) as minimumsalary from employees group by city having  
min(salary)>60000;
```

-- Q22=

```
select dept,city, count(name) as countemployees from employees group by dept,city;
```

-- Q23=

```
select dept,gender, count(name) as countemployees from employees group by  
dept,gender;
```

-- Q24=

```
select region,role, count(name) as countemployees from employees group by  
region,role order by region;
```

```
select state, dept , max(salary) as maxsalselect year(joindate) as joinyear, count(*)  
as joiner from emp group by joinyear order by joinyear; -- 30 ary from emp group  
by state,dept order by state; -- 25
```

select city, gender , sum(salary) as totalsalary from emp group by city, gender order by city; -- 26

select dept,city, count(empid) as empcount from emp group by dept,city having empcount > 2 order by dept;-- 27

select region, gender , avg(salary) as avgsalary from emp group by region, gender order by region, avgsalary desc; -- 28

select state, role , count(empid) as empcount from emp group by state, role having empcount > 1 order by state; -- 29

select year(joindate) as joinyear, count(*) as joiner from emp group by joinyear order by joinyear; -- 30

select year(joindate) as joinyear , avg(salary) as avgsalary from emp group by joinyear order by joinyear; -- 31

select month(joindate) as joinmonth , count(empid) as joiner from emp group by joinmonth order by joinmonth; -- 32

select quarter(joindate) as joinQtr , sum(salary) as totalsalary from emp group by joinqtr order by joinqtr; -- 33

select quarter(joindate) as Qtr , count(empid) as joiners from emp group by qtr having qtr in (1,3) ; -- 34

select year(joindate) as joinyear,dept, count(*) as empcount from emp group by joinyear,dept order by joinyear,dept; -- 35

```
select year(joindate) as joinyear, count(empid) as joiners from emp group by  
joinyear having joiners > 6 order by joinyear; -- 36
```

```
select quarter(joindate) as Qtr, avg(salary) as avgsalary from emp group by qtr  
having avgsalary > 70000 order by qtr; -- 37
```

```
select state, count(empid) as empcount ,avg(salary) as avgsalary from emp group  
by state order by state; -- 38
```

```
select state,dept, count(empid) as empcount from emp group by state,dept order  
by state; -- 39
```

```
select state,dept, count(empid) as empcount from emp group by state,dept order  
by state; -- 39
```

```
select city, count(dept) as deptcount from emp group by city having deptcount > 2; -  
- 40
```

```
select region,state , sum(salary) as totalsalary from emp group by region,state  
order by region; -- 41
```

```
select region,count(empid) as empcount , avg(salary) as avgsalary from emp group  
by region having empcount >10 and avgsalary > 67000; -- 42
```

```
select city,count(distinct(role)) as Uniqueroles from emp group by city order by city;  
-- 43
```

```
select dept, count(gender) as femalcount from emp where gender like 'f' group by  
dept order by dept; -- 44
```

```
select dept, avg(salary) as avgsalary from emp where joindate > '2019-01-01' group  
by dept order by dept; -- 45
```

```
select city, sum(salary) as totalsalary from emp where dept like 'it' group by city  
order by city; -- 46
```

```
select role, count(empid) as empcount from emp where salary > 70000 group by  
role order by empcount desc; -- 47
```

```
select dept, max(salary) as maxsalary from emp where salary > 70000 group by  
dept; -- 48  
select * from emp; select year(joindate) as joinyear, count(empid) as  
empcount from emp where state = 'tx' group by joinyear order by joinyear; -- 49
```

```
select dept, count(empid) as eligiblefemales from emp where gender = 'f' and  
salary > 60000 group by dept having count(empid) > 3; -- 50
```